

# Learning Resources

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All YouTube channels, video tutorials, online courses, articles, and documentation used during the development of Resilio. Organised by technology/topic. These represent the primary learning materials used to acquire or deepen understanding of each technology used in the project.

## FLUTTER AND DART

### YouTube — Primary Channels

| Channel                                                       | What Was Learned                                                                                                 | Key Videos / Playlists                                                                                        |
|---------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>**Reso Coder**</b> (resocoder.com / YouTube)               | Flutter BLoC pattern, Clean Architecture feature-first structure, Freezed, GetIt/Injectable dependency injection | "Flutter TDD Clean Architecture" course series; "Flutter BLoC Tutorial" series; "Freezed & json_serializable" |
| <b>**Flutter Official Channel**</b> (youtube.com/@flutterdev) | Dart fundamentals, widget lifecycle, Material Design 3, navigation patterns, performance                         | "Flutter Forward" talks; "Widget of the Week" series; "Flutter at Google I/O"                                 |
| <b>**Robert Brunhage**</b>                                    | Practical Flutter patterns, state management comparison                                                          | "Flutter BLoC vs Provider"; real-world project walkthroughs                                                   |
| <b>**Johannes Milke**</b>                                     | Widget-specific deep dives, animations                                                                           | Individual widget tutorials                                                                                   |
| <b>**Vandad Nahavandipoor**</b>                               | Advanced Dart, collections, async programming                                                                    | "Flutter Course for Beginners" (full playlist)                                                                |
| <b>**The Flutter Way**</b>                                    | Full project builds, UI implementation                                                                           | Full app builds for layout reference                                                                          |
| <b>**Code With Andrea**</b>                                   | Flutter architecture, testing patterns                                                                           | "Flutter & Firebase Masterclass"                                                                              |

### Key Topics Learned via YouTube

- Flutter BLoC (Business Logic Component) pattern — state management

- Cubit vs BLoC distinction — used Cubit for simpler screens, BLoC for complex flows
- GoRouter navigation with route guards
- Freezed code generation for immutable state models
- GetIt service locator and Injectable annotation processing
- flutter\_webrtc package setup and peer connection management
- sqflite offline SQLite integration
- Dio HTTP client with interceptors for auth token injection
- json\_serializable and code generation workflow (`flutter pub run build_runner build`)

## Official Flutter Documentation

- Flutter Docs: <https://docs.flutter.dev>
- Dart Docs: <https://dart.dev/guides>
- Flutter BLoC Library: <https://bloclibrary.dev>
- GoRouter: [https://pub.dev/packages/go\\_router](https://pub.dev/packages/go_router)
- Freezed: <https://pub.dev/packages/freezed>
- GetIt: [https://pub.dev/packages/get\\_it](https://pub.dev/packages/get_it)
- Injectable: <https://pub.dev/packages/injectable>
- flutter\_webrtc: [https://pub.dev/packages/flutter\\_webrtc](https://pub.dev/packages/flutter_webrtc)
- sqflite: <https://pub.dev/packages/sqflite>
- Dio: <https://pub.dev/packages/dio>

## SUPABASE (PostgreSQL + RLS)

### YouTube

| Channel                                                                                            | What Was Learned                                                                                |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| <b>**Supabase Official**</b> ( <a href="https://youtube.com/@Supabase">youtube.com/@Supabase</a> ) | Row Level Security setup; Supabase client in Dart; database migrations; real-time subscriptions |
| <b>**Fireship**</b>                                                                                | "Supabase in 100 seconds"; SQL fundamentals refresher                                           |
| <b>**Jon Meyers**</b>                                                                              | "Build a full-stack app with Supabase and Flutter"                                              |

## Key Topics Learned

- PostgreSQL table creation with foreign keys and constraints

- Row Level Security (RLS): writing policies with `auth.uid()` and role checks
- Supabase JavaScript and Dart clients
- Supabase dashboard for table management and SQL editor
- Real-time listeners for appointment status changes
- Storage buckets (used briefly before switching to Cloudinary)

## Official Documentation

- Supabase Docs: <https://supabase.com/docs>
- Supabase Row Level Security Guide: <https://supabase.com/docs/guides/auth/row-level-security>
- Supabase Flutter Client: <https://supabase.com/docs/reference/dart/introduction>
- PostgreSQL Docs: <https://www.postgresql.org/docs/>

## NODE.JS AND EXPRESS

### YouTube

| Channel                                         | What Was Learned                                                         |
|-------------------------------------------------|--------------------------------------------------------------------------|
| <b>**Traversy Media**</b>                       | Express.js crash course; REST API with Node.js; middleware patterns      |
| <b>**The Net Ninja**</b>                        | Node.js full course; JWT auth; async/await patterns in Express           |
| <b>**Academind**</b> (Maximilian Schwarzmüller) | REST API design; error handling middleware; modular Express architecture |
| <b>**Fireship**</b>                             | "Node.js in 100 seconds"; "Express.js in 100 seconds"                    |
| <b>**Hussein Nasser**</b>                       | Backend fundamentals; database connection pooling; HTTPS setup           |

### Key Topics Learned

- Express.js routing and middleware
- Authentication middleware (validating SuperTokens session tokens)
- Repository pattern for data layer separation
- Environment variable management with dotenv
- CORS configuration for mobile clients
- Error handling middleware patterns

- Async/await with try-catch in route handlers

## Official Documentation

- Node.js Docs: <https://nodejs.org/en/docs/>
- Express.js Docs: <https://expressjs.com/en/4x/api.html>
- npm: <https://docs.npmjs.com/>

## SUPERTOKENS (Authentication)

### YouTube / Tutorials

| Resource                                                                                                 | What Was Learned                                                                                                 |
|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <b>**SuperTokens Official Blog**</b> ( <a href="https://supertokens.com/blog">supertokens.com/blog</a> ) | Passwordless recipe setup; session management; FDI protocol                                                      |
| <b>**SuperTokens GitHub Examples**</b>                                                                   | Node.js backend init; Flutter frontend integration patterns                                                      |
| Searched: "SuperTokens passwordless Flutter" — various dev.to and Medium articles                        | Three-step OTP flow: <code>createCode</code> → <code>consumeCode</code> → custom <code>/complete</code> endpoint |

### Key Topics Learned

- SuperTokens self-hosted vs managed service (used managed)
- Passwordless recipe: `createCode`, `consumeCode` API calls
- FDI (Frontend Driver Interface) — the protocol the Flutter client uses
- Session token handling: `st-access-token` returned in response headers
- Google Sign-In and Facebook Sign-In provider setup in SuperTokens dashboard
- Custom `/auth/passwordless/complete` backend endpoint to sync user into Supabase after auth

## Official Documentation

- SuperTokens Docs (Passwordless): <https://supertokens.com/docs/passwordless/introduction>
- SuperTokens Node.js SDK: <https://supertokens.com/docs/nodejs/quickstart>
- SuperTokens FDI Spec: <https://supertokens.com/docs/contribute/apis/fdi>

## WEBRTC AND SOCKET.IO

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### YouTube

| Channel / Video                             | What Was Learned                                                             |
|---------------------------------------------|------------------------------------------------------------------------------|
| **Fireship** — "WebRTC in 100 seconds"      | WebRTC fundamentals: SDP, ICE candidates, STUN/TURN                          |
| **Fireship** — "WebRTC Crash Course" (full) | Full peer connection lifecycle; signalling architecture                      |
| **Hussein Nasser**                          | "How WebRTC Works" — deep dive into ICE, STUN, TURN; NAT traversal           |
| **The Net Ninja** — Socket.IO series        | Socket.IO rooms; emit/on patterns; namespace vs room; disconnection handling |
| **WebRTC.org official samples**             | flutter_webrtc package integration reference                                 |
| Various Medium/dev.to articles              | "Flutter WebRTC video call with Socket.IO signalling" implementation guides  |

### Key Topics Learned

- **SDP (Session Description Protocol):** offer/answer exchange between peers
- **ICE (Interactive Connectivity Establishment):** candidate gathering; STUN for public IP discovery
- **TURN server:** relay server when direct peer connection fails; deployed for fallback
- **Socket.IO signalling flow:** join-room → offer → answer → ice-candidate → leave-room
- **flutter\_webrtc RTCPeerConnection:** creating offers, setting remote descriptions, adding ICE candidates
- **Media streams:** getUserMedia for camera/microphone access in Flutter
- **Socket.IO rooms:** how both client and therapist join the same meeting room ID

### Official Documentation

- WebRTC Official: <https://webrtc.org>
- flutter\_webrtc GitHub: <https://github.com/flutter-webrtc/flutter-webrtc>
- Socket.IO Docs v4: <https://socket.io/docs/v4>
- STUN/TURN: <https://webrtc.org/getting-started/turn-server>

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## FIREBASE (FCM Push Notifications)

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## YouTube

| Channel                                                       | What Was Learned                                                        |
|---------------------------------------------------------------|-------------------------------------------------------------------------|
| <b>**Firebase Official**</b> (youtube.com/@Firebase)          | FCM setup; Flutter Firebase integration; notification payload structure |
| <b>**Reso Coder**</b> — "Flutter Firebase Push Notifications" | flutter_local_notifications integration; foreground/background handling |
| <b>**The Net Ninja**</b> — Firebase series                    | Firebase project setup; google-services.json configuration              |

## Key Topics Learned

- FCM token registration and storage per user in Supabase
- Sending notifications from Node.js backend using firebase-admin SDK
- Handling foreground, background, and terminated app notification states in Flutter
- Notification payload structure: title, body, data fields
- Deep linking from notification to specific screen in app

## Official Documentation

- Firebase Flutter Setup: <https://firebase.flutter.dev/docs/overview>
- FCM (Flutter): <https://firebase.flutter.dev/docs/messaging/overview>
- firebase-admin (Node.js): <https://firebase.google.com/docs/admin/setup>

## ESEWA PAYMENT INTEGRATION

### Resources Used

| Resource                                   | What Was Learned                                             |
|--------------------------------------------|--------------------------------------------------------------|
| <b>**eSewa Developer Portal**</b>          | API documentation; test credentials; callback verification   |
| <b>**eSewa Flutter SDK GitHub**</b>        | SDK initiation; ESewaConfig, ESewaPayment, callback handlers |
| <b>**fonepay.com docs**</b> (eSewa parent) | HMAC signature verification for server-side validation       |

| Resource                                                   | What Was Learned                                             |
|------------------------------------------------------------|--------------------------------------------------------------|
| Nepali developer community (GitHub issues, Stack Overflow) | Debugging payment callback URL and SDK version compatibility |

## Key Topics Learned

- eSewa test environment vs production environment setup
- ESewaFlutterSdk.initPayment() parameters: productId, productName, productPrice
- Three callback handlers: onPaymentSuccess, onPaymentFailure, onPaymentCancellation
- Server-side HMAC signature verification to prevent tampered payment callbacks
- Storing transaction reference IDs for audit trail

## Official Documentation

- eSewa Flutter SDK: [https://pub.dev/packages/esewa\\_flutter\\_sdk](https://pub.dev/packages/esewa_flutter_sdk)
- eSewa Developer Portal: <https://developer.esewa.com.np>

## CLOUDINARY (Media Storage and Delivery)

### Resources Used

| Resource                                     | What Was Learned                                          |
|----------------------------------------------|-----------------------------------------------------------|
| **Cloudinary Documentation**                 | Upload API; transformation URLs; unsigned upload presets  |
| YouTube: "Cloudinary with Flutter" tutorials | flutter_cloudinary_public package; image and video upload |
| Cloudinary blog                              | CDN delivery; adaptive bitrate streaming for video        |

## Official Documentation

- Cloudinary Docs: <https://cloudinary.com/documentation>
- Upload API: [https://cloudinary.com/documentation/image\\_upload\\_api\\_reference](https://cloudinary.com/documentation/image_upload_api_reference)

## PROTOCOL BUFFERS (Offline Serialisation)

## Resources Used

| Resource                                              | What Was Learned                                     |
|-------------------------------------------------------|------------------------------------------------------|
| Protocol Buffers official docs                        | .proto file syntax; generating Dart code from .proto |
| YouTube: "Protocol Buffers in 100 seconds" (Fireship) | Why protobuf over JSON: binary efficiency            |
| protobuf Dart package docs                            | Encoding/decoding binary messages in Flutter         |

## Official Documentation

- Protocol Buffers: <https://protobuf.dev>
- protobuf Dart: <https://pub.dev/packages/protobuf>

## GENERAL SOFTWARE ENGINEERING

### YouTube

| Channel            | What Was Learned                                                       |
|--------------------|------------------------------------------------------------------------|
| **Fireship**       | Technology overviews; quick conceptual primers for many tech decisions |
| **Traversy Media** | Full-stack project patterns; REST API fundamentals                     |
| **Tech With Tim**  | Python scripting for data processing (pre-survey analysis)             |
| **Academind**      | Clean code; modular architecture concepts                              |

## Books and Written Resources

| Resource                                                  | Used For                                                      |
|-----------------------------------------------------------|---------------------------------------------------------------|
| *Clean Architecture* — Robert C. Martin (2017)            | Feature-first architecture design; layer separation rationale |
| *The Rational Unified Process* — Philippe Kruchten (2003) | RUP methodology framework; phase definitions                  |



| Resource                                                                  | Used For                                                   |
|---------------------------------------------------------------------------|------------------------------------------------------------|
| *Software Engineering* — Ian Sommerville (2016)                           | Testing frameworks; non-functional requirements categories |
| *Software Engineering: A Practitioner's Approach* — Roger Pressman (2014) | System testing methodology; test case design               |

## DESIGN TOOLS

| Tool                       | Used For                              | Learning Resource                                       |
|----------------------------|---------------------------------------|---------------------------------------------------------|
| **Figma**                  | UI mockup design; component library   | Figma official YouTube; "Figma for Beginners" tutorials |
| **draw.io / diagrams.net** | UML diagrams, ERD, data flow diagrams | Built-in documentation; YouTube tutorials               |
| **Mermaid**                | Inline markdown UML in report         | mermaid.js.org official docs; Live Editor               |
| **Lucidchart**             | Gantt chart and WBS creation          | Lucidchart tutorial videos                              |

## ADDITIONAL REFERENCE ARTICLES (Dev.to / Medium / Stack Overflow)

| Topic                                | Platform                        | Description                                             |
|--------------------------------------|---------------------------------|---------------------------------------------------------|
| Flutter BLoC with Clean Architecture | Medium (Reso Coder)             | Detailed walkthrough of feature-first file structure    |
| SuperTokens + Flutter integration    | dev.to                          | Community article: integrating the FDI API from Flutter |
| WebRTC with Socket.IO in Flutter     | Medium                          | Step-by-step signalling implementation                  |
| Supabase RLS policies in Flutter     | dev.to                          | Writing row-level security for multi-role apps          |
| eSewa integration Nepal              | GitHub Gists / Nepali dev blogs | Community guidance on SDK usage and HMAC verification   |

| Topic                              | Platform      | Description                                         |
|------------------------------------|---------------|-----------------------------------------------------|
| flutter_webrtc STUN configuration  | GitHub Issues | Resolving ICE candidate failures on mobile networks |
| Protobuf with Flutter offline sync | Medium        | Binary serialisation for local SQLite queues        |